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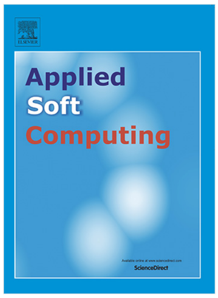


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PII: S1568-4946(20)30433-6
DOI: https://doi.org/10.1016/j.asoc.2020.106494
Reference: ASOC 106494

To appear in: *Applied Soft Computing Journal*

Received date : 10 October 2019
Revised date : 18 May 2020
Accepted date : 16 June 2020

Please cite this article as: Balaji E., Brindha D. and Balakrishnan R., Supervised machine learning based gait classification system for early detection and stage classification of Parkinson's disease, *Applied Soft Computing Journal* (2020), doi: https://doi.org/10.1016/j.asoc.2020.106494.

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# Supervised Machine Learning based Gait Classification System for Early Detection and Stage Classification of Parkinson's Disease

Balaji E[a,*], Brindha D[a], Balakrishnan R[b]

[a]*Department of Biomedical Engineering, PSG College of Technology, Coimbatore, India - 641004*
[b]*Department of Neurology, PSG Insitute of Medical Sciences & Research, Coimbatore, India - 641004*

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**Abstract**

While diagnosing Parkinson's disease (PD), neurologists often use several clinical manifestations of the subject and rate the severity level based on the Unified Parkinson Disease Rating Scale (UPDRS). This kind of rating largely depends on the expertise of the doctors, which is not only subjective but also inefficient. Hence, in this paper, a machine learning based gait classification system which can assist the clinician to diagnose the stages of PD is presented. Gait pattern, which plays a significant role in assessing the human mobility, is a significant biomarker to classify whether the subject is healthy or affected with PD. Hence, we utilize the vertical ground reaction force (VGRF) gait dataset and extract the minimal feature vector using the statistical analysis. Subsequently, the normal distribution of the data is validated using the Shapiro-Wilk test, and from the spatial and temporal features of gait pattern, the salient biomarkers are identified using the correlation based feature selection technique. Four supervised machine learning algorithms namely decision tree (DT), support vector machine (SVM), ensemble classifier (EC) and Bayes classifier (BC) are used for statistical and kinematic analyses which predict the severity of PD. The classifier efficacy quantified using the accuracy, sensitivity and specificity highlights that the proposed framework can effectively rate the severity of PD based on Hohen and Yahr (H&Y) scale. Moreover, comparing the accuracy of the proposed PD classification approach with those of the other state-of-the-art approaches, which utilized the same gait dataset, reveal that the proposed method outperforms several other PD classification methods.

*Keywords:* Parkinson's disease (PD), Machine learning algorithms, Classifiers, Confusion matrix, Temporal and spatial features, H& Y scale.

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## 1. Introduction

Parkinson's disease is a chronic neurodegenerative brain disorder, which affects the ability of the person to perform the regular activities. Parkinson's foundation (PF), USA estimates that around 8-10 million people are worldwide affected by the PD, which is considered to be the second most common neurodegenerative disorder next to Alzheimer's disease [1]. Arvid Carlsson, a Swedish neuroscientist, who won Nobel prize in Physiology in 2000, identified that the major reason for PD is due to the lack of a chemical messenger in the brain called Dopamine. Dopamine, which is present in the specific area of a brain called substantia nigra, is a neurotransmitter that not only controls the motor activity but coordinates the muscle functions. Numerous non-motor symptoms of PD start ahead of the motor symptoms [2] [3]. Three of the most common motor symptoms of PD are: tremor, limb rigidity, and slowness of movement (bradykinesia). Similarly, the common non-motor symptoms of PD are sleep disorder, depression and fatigue. Person affected with PD may also have problem with balance, posture and coordination. Even though the symptoms of PD are common, progression of PD may vary largely from person to person and change over time [4] [5].

The popular methods available to diagnose PD are finger motion analysis [6], functional neuroimaging [7], surface EMG analysis [8], and foot pressure analysis [9]. The conventional diagnosis largely depends on the subjective measures obtained from visual observations and questionnaire of the clinicians to prepare a score based on UPDRS, which consists of 42 questions to assess the motor symptoms, daily activities and behavioral characteristics. When clinicians utilize these scales to classify the severity level of PD, chances of misclassification and low efficiency are ineluctable because several of the diagnostic criteria are descriptive symptoms, which fail to provide a quantified diagnostic basis [10]. Hence, recently, significant interest has been shown on machine learning (ML) algorithms to assist the clinicians in their daily diagnosis of PD [11]. One of the major reasons for the paradigm shift towards ML solutions is their capability to handle large volume of heterogeneous clinical and imaging data and identify the hidden pattern in the dataset. In literature, the potential applications of ML algorithms to diagnose PD based on the pathologi-

*Corresponding author

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cal speech signals [12], motion signals [13], handwriting dynamics [14], and gait signals [15] are considerably studied and the idea of automating the PD diagnosis is being translated into a reality in clinical practice [16].

Interestingly, the tremor and gait abnormality are the initial manifestations of PD. Gait pattern, which involves the sequence of periodic and rhythmical pattern of foot movements, has some interesting and distinct properties like periodicity, irregular cycle and deterministic behavior [17][18]. Particularly, the kinematic features of gait pattern have significant biomarkers which can be critical for not only identifying the presence of PD but also quantifying the progression of the disease [19]. Moreover, the spatiotemporal variables are less prone to physiological parameter like age, height and weight; hence, they can serve as the basis to extract the significant features for PD prognosis. Even though the PD diagnosis has been studied extensively based on gait cycle analysis, the kinematic features of gait abnormality for prognosis of PD have not been investigated to its full potential. Hence, the major objective of this paper is to explore the significant spatiotemporal features of the gait pattern so that the stages of PD, which need finer investigation of walking pattern, can be precisely identified.

In this study, an automatic and non-invasive gait classification system for PD diagnosis and severity rating is presented. From the gait pattern acquired using 16 foot worn sensors, the prominent temporal and spatial features are extracted. To obtain the optimal feature set for the classification model, a correlation based feature selection technique is employed. Firstly, the statistical analysis of the VGRF sensor data is performed to differentiate between healthy control and PD patients. In statistical analysis, the weight of the subject influences the foot plantar pressure and classifier output. Hence, we also perform kinematic analysis to avoid biased classification. Unlike several other ML based approaches which perform the binary classification, in this work, we aim to solve the multi-class classification problem of PD using discriminative feature set obtained from the spatiotemporal features. Four supervised machine learning algorithms namely decision tree(DT), support vector machine (SVM), ensemble classifier(EC) and Bayes classifier are used to classify the stages of PD based on H & Y scale. Moreover, to avoid data overfitting problem and enhance the classification accuracy, the 10 fold cross validation technique is utilized. The efficacy of the classifier model is validated using the region of convergence (ROC) curve and confusion

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matrix.

The remainder of the paper is organized as follows. Section 2 presents a review of previous works on machine learning based gait classification for PD diagnosis. Section 3 briefs the VGRF dataset and the demographic details of the subjects considered for the study. Section 4 discusses the proposed gait classification approach along with the feature extraction techniques and supervised ML classifier algorithms. Section 5 introduces the metrics used for assessing the classifier performance. Section 6 presents the results and discussion of both statistical and kinematic analyses of PD classification. Finally, section 7 gives the concluding remarks of the paper.

## 2. Related Work

Several previous studies have assessed the gait features of subjects affected with Parkinsonism. For instance, based on gait and tremor features extracted using statistical analysis, Perumal and Sankar [20] presented a linear discriminant analysis (LDA) based pattern classification algorithm for early detection and monitoring of PD. Utilizing the kinetic features to identify a PD tremor due to Parkinsonism, they analysed the frequency domain characteristics of tremor and obtained an average accuracy of 86.9%. However, their approach is reduced to a binary classification problem in which only the presence of PD is detected and stages of PD have not been identified. Aşuroğlu et al. [21] proposed a locally weighted random forest regression model to eliminate the effects of interpatient variability in gait features and utilized the ground reaction force sensor data to model the relationship between gait patterns and PD symptoms. Based on the 16 time-domain and 7 frequency-domain features, they provided a quantitative assessment of PD motor symptoms. Nevertheless, only the statistical analysis of GRF, which is influenced by the weight of the subject, is used for identifying the PD symptoms and the kinematic analysis and the severity level of PD have not been reported.

Using wavelet based feature extraction technique for gait characteristics and extracting 40 features based on statistical analysis and frequency distributions, Lee and Lim [22] proposed a PD classification approach which made use of neural network with weighted fuzzy membership functions and obtained the maximum classification accuracy of 81.63%. To differentiate between

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idiopathic PD patients and healthy subjects, they used the bounded sum of weighted fuzzy membership functions for selecting minimum features based on a non-overlapping area distribution method. Nonetheless, the large number of gait features obtained from the wavelet approximation and detailed coefficients will significantly increase the computational complexity. For computer assisted PD diagnosis and severity detection, Zhao et al. [10] put forward a two channel model, which fuses long short-term memory and convolutional neural network to learn the spatiotemporal gait pattern from VGRF data. Even though significant improvement in the classifier accuracy is achieved using the deep learning architecture, it has been reported that higher prediction accuracy can be achieved by fusing the multiple data sources like MRI image and biochemical data.

To understand the periodicity and randomness in gait signals, Prabhu et al., [23] performed the recurrence quantification analysis (RQA) along with statistical analysis and reported that RQA works well even for short length gait time series data. Moreover, for optimal selection of feature vectors, they used Hill-climbing feature selection method and compared the classification performance of probabilistic neural network with that of the SVM. Figueiredo et al., [24] showed that using kernel principal component analysis and genetic algorithm, which have the capability to process the nonlinear data and converge to global optimum, could significantly reduce the dimensionality features of gait parameters for efficient classification. They presented a comparative analysis of machine learning approaches for gait pattern recognition, and used a cross validation method to assess the performance of classification algorithms for human walking recognition in clinical applications.

Oung et al. [25] proposed a multi-class PD classification approach with the combination of speech signals and gait pattern, which are analysed based on empirical wavelet transform (EWT) and empirical wavelet packet transform (EMPT), respectively. Applying the Hilbert transform to extract the wavelet energy and entropy based features, they achieved a classification accuracy of 95% using the extreme learning machine classifier. However, the feature dimensionality selection/reduction to obtain the optimal feature set, which can reduce the computational burden, has not been implemented. To obtain the significant differences in the spatial-temporal features between PD patients and healthy controls, Wahid et al.[26] used a multiple regression (MR) normalization approach and evaluated the efficacy of machine learning algorithms in classifying PD gait after normalization. Notwithstanding, the MR model was developed and tested for a relatively small dataset with body mass index of majority of the subjects being greater than 25 that may produce increased dispersion of data.

Several of the aforementioned studies have used time-domain and frequency-domain features to differentiate the healthy control and PD subjects. Nevertheless, the major limitation is that such features can not be directly related to clinical indicators which could assist the neurologists to rate the severity level [27, 15]. Therefore, the prime focus of this paper is to develop a functional tool for diagnosing and severity rating of PD based on the VGRF data and seek to identify the discriminative kinematic features which can be linked to the clinical indicators. Furthermore, using the correlation based optimal feature set extracted from spatiotemporal domain, the proposed PD diagnosis approach aims to improve the classifier accuracy.

## 3. Dataset Description

In this study, the public dataset for gait pattern provided by Physionet is used [28]. The dataset consists of 279 gait recordings from 73 healthy subjects (40 men and 33 women, mean age: 63.7 years) and 93 idiopathic PD patients (58 men and 35 women, mean age: 66.3 years). This dataset originally collected by the Laboratory for Gait & Neurodynamics, Movement Disorders Unit of the Tel-Aviv Sourasaky Medical Centre, includes measures of different levels of disease severity based on the H & Y and/or the UPDRS rating [29]. The database contains 10 PD patients with H & Y scale=3, 28 PD patients with H & Y scale=2.5, and 55 PD patients with H & Y scale=2. Hence, the dataset indicates that most of the subjects were at the early stage of the disease or with moderate severity, which can serve as a benchmark to assess the proposed stage identification of PD using statistical and kinematic analyses [30]. Table I gives the demographic details of the VGRF dataset. For data collection, 8 VGRF sensors were placed in the insole of each of the subject's foot, and each subject was asked to walk at their usual, self-styled pace for 2 minutes on level ground. Hence, each recording comprises 16 VGRF time series data in addition to 2 time series data that give the cumulative force of each foot. The VGRF signals measured in Newtons were sampled at a rate of 100 Hz. Figure 1 illustrates the positioning of VGRF sensors in left and right feet to capture the gait pattern. Table II gives the relative position of the 16 sensors in X-Y



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coordinate system, assuming that the origin is at (0,0) when each leg is facing towards the positive side of the Y-axis.

Table I: Demographic features of healthy control and PD patients

| Group | Subjects | | Age (Yrs) | | Weight (kg) | | Height (m) |
|---|---|---|---|---|---|---|---|
| | Male | Female | Male | Female | Male | Female | |
| Healthy | 40 | 33 | 63.87±42.16 | 63.47±38.07 | 64.69± 25.57 | 79.10±19.08 | 1.68±.08 |
| Mild | 36 | 19 | 61.6± 19.01 | 65.3± 11.42 | 64.66± 17.14 | 78.67 ± 9.44 | 1.67±.06 |
| Medium | 16 | 12 | 69.9±10.39 | 68.6± 11.13 | 66.75± 8.53 | 79.9± 7.13 | 1.71±0.07 |
| High | 6 | 4 | 64.2 ± 11.58 | 75±11.20 | 59.75± 6.29 | 67.2±6.61 | 1.68±.06 |

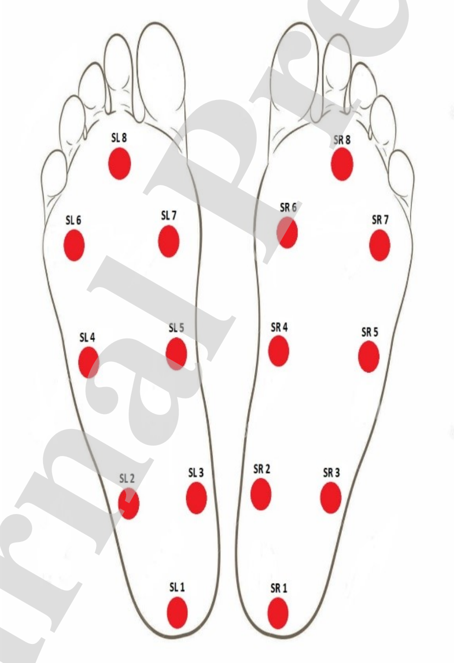


Figure 1: Positioning of VGRF sensors

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Table II: Relative position of sensors in left and right feet

| Sensor | X axis (mm) | Y axis (mm) |
|---|---|---|
| SL1 | -500 | -800 |
| SL2 | -700 | -400 |
| SL3 | -300 | -400 |
| SL4 | -700 | 0 |
| SL5 | -300 | 0 |
| SL6 | -700 | 400 |
| SL7 | -300 | 400 |
| SL8 | -500 | 800 |
| SR1 | 500 | -800 |
| SR2 | 700 | -400 |
| SR3 | 300 | -400 |
| SR4 | 700 | 0 |
| SR5 | 300 | 0 |
| SR6 | 700 | 400 |
| SR7 | 300 | 400 |
| SR8 | 500 | 800 |

## 4. Methodology

Figure 2 shows the proposed gait classification framework for severity prediction of PD. In data analysis, to reduce the start-up and end-up effects of gait, first and last two gait cycles of each trial were discarded. The statistical and kinematic analyses of the gait pattern are individually performed to extract the prominent biomarkers. Subsequently, the optimal feature set obtained using the rank correlation method is given to the classifier model to predict the stages of PD [31]. To assess the performance of the classifier model 6 performance metrics are reported. In the following section, we present the statistical and kinematic feature extraction techniques and discuss the ML algorithms used in the classifier model.

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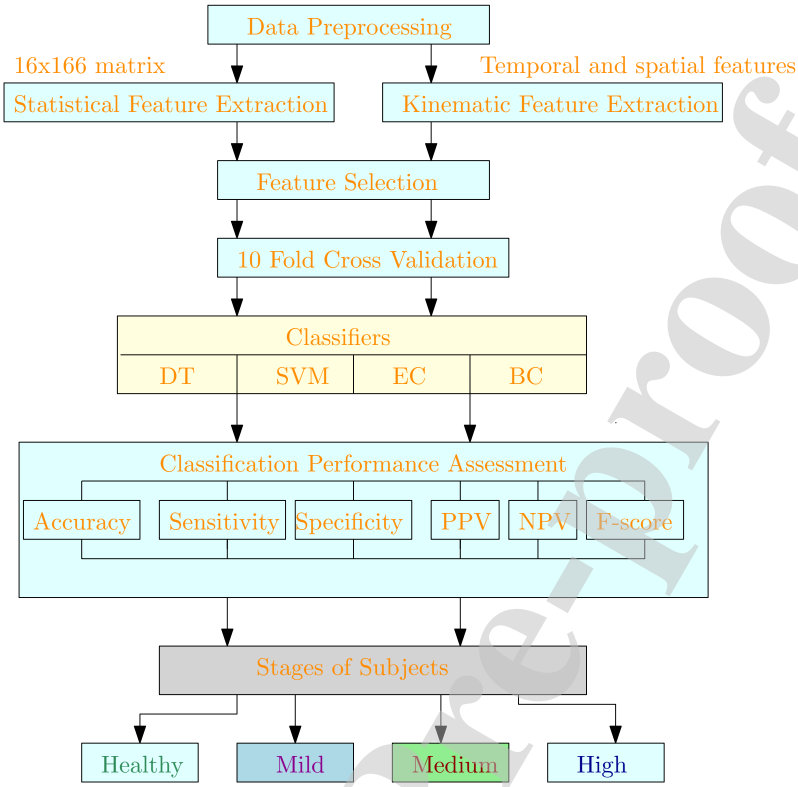


Figure 2: Proposed gait classification framework for stage classification of PD

### 4.1. *Feature Extraction using statistical analysis*

We compute the fluctuation magnitudes of 16 sensors to statistically analyse the gait pattern of both healthy subjects and PD patients. Table III, which gives the mean, median and standard deviation of both the subjects, reveals that the fluctuation of the mean value of the respective sensors in 2 feet of the PD patient is relatively higher than that of the healthy subjects, particularly in SL1, SL4, SL5 and SL6 for left feet sensors and SR1, and SR7 for right feet sensors. Due to higher fluctuations of mean value, we utilize the mean of the VGRF sensor data as a significant biomarker for PD classification. Moreover, for assessing the efficacy of the classifiers, we use 10 fold cross validation approach, which divides the dataset into 10 subsets. Each subset is cross-validated with other subsets such that 90% of data is used for training and 10% of data is utilized for testing. As a result, it avoids data overfitting and guarantees that the already trained and tested data will not be included for the next subset. Furthermore, by applying the Shapiro-Wilk test with the significance level of 5%, the normal distribution of the gait attributes is assessed. Figure 3 illustrates the foot pressure output of VGRF sensor for both healthy and PD subjects, and Figure 4 shows the input matrix obtained from 16 sensors for the 166 subjects based on the H & Y scale for the four stages of PD. Since the no of samples in the early stages of PD is significantly high, it can assist to reasonably capture the relationship between the input and output feature sets.

Table III: Statistical parameters of VGRF signals from 16 sensors

| Sensors | Healthy subjects | | | PD patients | | |
|---|---|---|---|---|---|---|
| | Mean | Median | SD | Mean | Median | SD |
| SL1 | 77.71 | 85.80 | 29.98 | 56.54 | 50.29 | 32.45 |
| SL2 | 63.66 | 63.35 | 20.43 | 55.15 | 51.56 | 20.42 |
| SL3 | 51.52 | 48.43 | 20.47 | 51.17 | 48.97 | 21.17 |
| SL4 | 57.56 | 51.94 | 23.42 | 66.28 | 63.34 | 25.89 |
| SL5 | 25.54 | 18.29 | 21.08 | 38.89 | 33.76 | 26.67 |
| SL6 | 66.61 | 64.60 | 20.72 | 67.34 | 67.42 | 21.29 |
| SL7 | 71.71 | 68.58 | 20.16 | 78.89 | 78.73 | 22.50 |
| SL8 | 36.67 | 32.35 | 19.91 | 38.99 | 33.98 | 21.35 |
| SR1 | 75.48 | 79.62 | 28.22 | 55.50 | 56.73 | 30.91 |
| SR2 | 62.74 | 59.37 | 19.69 | 55.45 | 53.02 | 19.26 |
| SR3 | 54.14 | 52.65 | 23.02 | 52.95 | 49.79 | 22.94 |
| SR4 | 61.93 | 59.98 | 22.12 | 67.77 | 67.87 | 21.88 |
| SR5 | 30.40 | 23.12 | 23.76 | 36.62 | 31.02 | 23.15 |
| SR6 | 65.31 | 64.35 | 22.08 | 65.66 | 66.16 | 19.01 |
| SR7 | 71.39 | 66.94 | 22.12 | 80.66 | 78.41 | 22.24 |
| SR8 | 39.29 | 34.94 | 17.44 | 39.09 | 37.31 | 14.24 |

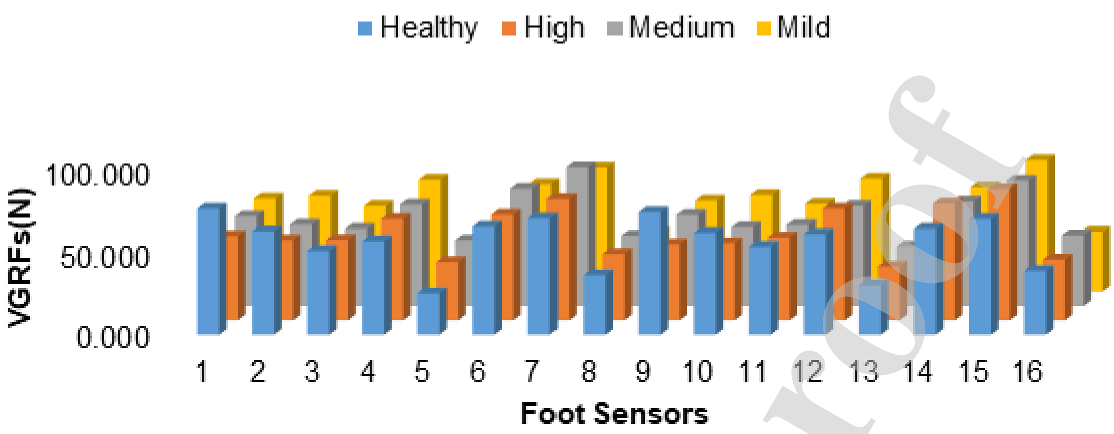


Figure 3: Mean plantar force of 16 sensors

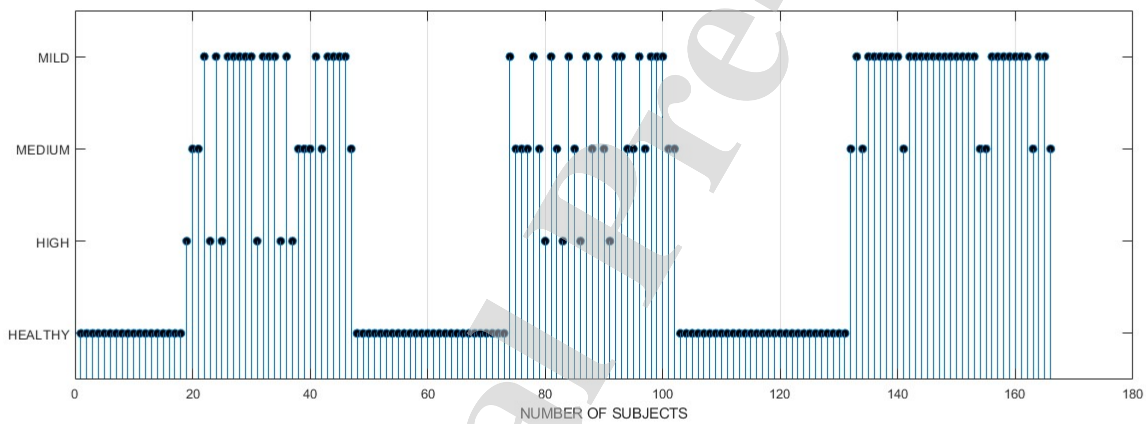


Figure 4: Classifier input pattern (16x166 matrix) with stages

### 4.2. *Kinematic analysis of gait pattern*

The kinematic gait analysis helps to assess progression of PD. PD affects the regular walking pattern of the subject and drastically reduces the number of footsteps, speed, step time, and stride interval. Hence, the study of gait cycle parameters such as step time, stride time, stance time, swing time, swing stance ratio, cadence, speed, step length and stride length, which act as significant biomarkers, help to understand the gait pattern disorder of PD subjects. Since feature selection plays a significant role in identifying the most discriminative features for efficient classification model, in this study, a correlation based feature selection technique is used to identify a set of informative and prominent gait attributes for efficient stage classification. Figure 5 illustrates various gait parameters of a subject during normal walk. The gait pattern contains 62% of stance interval and 38% of swing interval. In this work, the gait parameters are split into two components namely temporal and spatial features, which are briefed below.

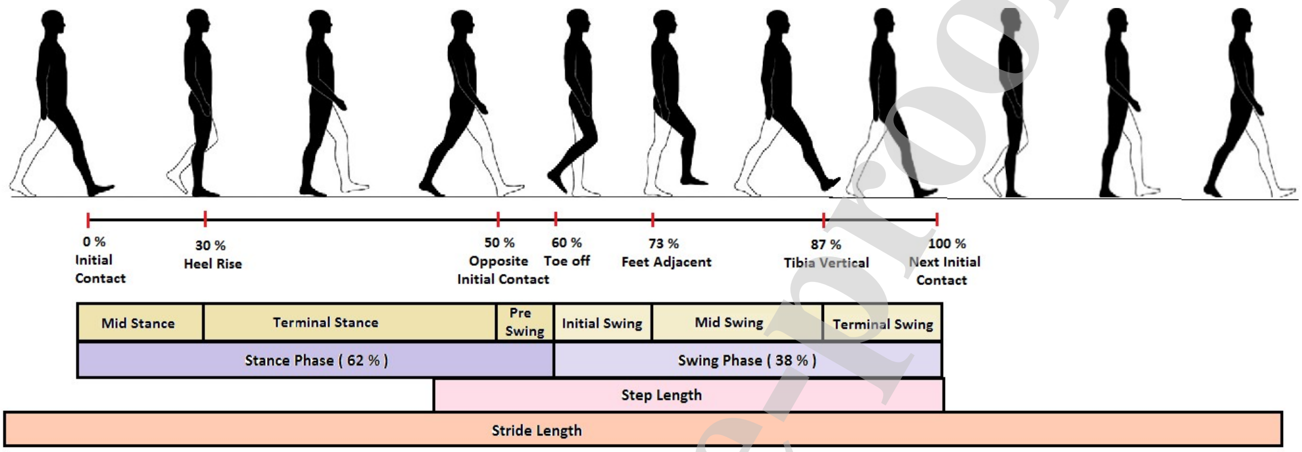


Figure 5: Different phase representation of gait pattern

#### *4.2.1.* ***Temporal features***

**Step time:**It is the time duration between heel strike of one leg and heel contact of the opposite leg. The approximate step time of healthy subject is 0.8 s.

**Stance time:** It is the time interval between the contact of the heel to the surface and the contact of the opposite leg when the toes do not touch the surface. Stance phase starts with the heel strike, and the mid stance is a complete body weight balance in one leg when another leg is in swing phase.

**Swing time:** Swing time is observed between the heel strike and toe off phase. Swing phase starts at the end of stance.

**Swing stance ratio (SS ratio):** It is a ratio between the swing time and stance time.

**Stride time**: It is the time duration between foot ground contact of successive instants of same foot.

**Cadence:** It is the total number of steps/unit time. The nominal values of cadence are: slow :60-70 steps/min, medium: 70-90 steps/min, and normal:90-115 steps/min.

**Speed:** The total distance covered / unit time gives the speed, and it is measured in m/sec. Typical values for male and female are 80 m/min, and 70 m/min, respectively.

#### *4.2.2. **Spatial features***

**Step length:** It is the distance between the successive footsteps of heel contact of two foot steps. Typically, the right foot step length is equal to left foot step length.

**Stride length:** It is the distance between the successive foot of heel contact of the same foot, and it is also called as double step length.

Table IV gives the 9 temporal and spatial features of 93 PD and 73 healthy subjects based on the sum of left and right feet sensor values. It can be noted that the swing time remains almost same for different stages of PD, whereas the stance time has relatively considerable fluctuations. The highest variation can be observed in the features such as cadence, stride length, stride time and speed. Figure 6 shows the temporal and spatial features of both healthy and PD subjects for different stages. The spatiotemporal features highlight that the cadence is the most influential factor which has larger variability due to the severity of PD. Even though the mild and medium stages of PD have similar cadence, compared to cadence of healthy control, which is 110.325, subjects with high severity of PD exhibit drastic reduction in cadence which is around 101.625. Similarly, the second most significant feature is the stride length, which reads for healthy-1.35, high-0.939, medium-1.115 and the mild-1.2.

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Table IV: Temporal and spatial features of PD and healthy subjects

| Spatiotemporal features | Healthy | | Mild | | Medium | | High | |
|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | Mean | SD |
| Swing Time (s) | 0.399 | ±0.034 | 0.389 | ±0.051 | 0.378 | ±0.027 | 0.399 | ±0.041 |
| Stance Time (s) | 0.696 | ±0.060 | 0.847 | ±0.239 | 0.736 | ±0.083 | 0.713 | ±0.058 |
| Stride Time (s) | 1.095 | ±0.088 | 1.235 | ±0.269 | 1.114 | ±0.099 | 1.112 | ±0.087 |
| SS Ratio (s) | 0.575 | ±0.038 | 0.481 | ± 0.092 | 0.518 | ±0.053 | 0.561 | ±0.051 |
| Step Time (s) | 0.547 | ±0.044 | 0.618 | ±0.135 | 0.557 | ±0.049 | 0.556 | ±0.043 |
| Step Length (m) | 0.675 | ±0.068 | 0.469 | ±0.124 | 0.558 | ±0.098 | 0.600 | ±0.095 |
| Stride Length (m) | 1.35 | ±0.137 | 0.939 | ±0.248 | 1.115 | ±0.196 | 1.200 | ±0.191 |
| Speed (m/s) | 1.241 | ±0.160 | 0.808 | ±0.307 | 1.008 | ±0.183 | 1.083 | ±0.169 |
| Cadence (steps/min) | 110.325 | ±8.847 | 101.625 | ±21.521 | 108.481 | ±9.200 | 108.625 | ±9.006 |

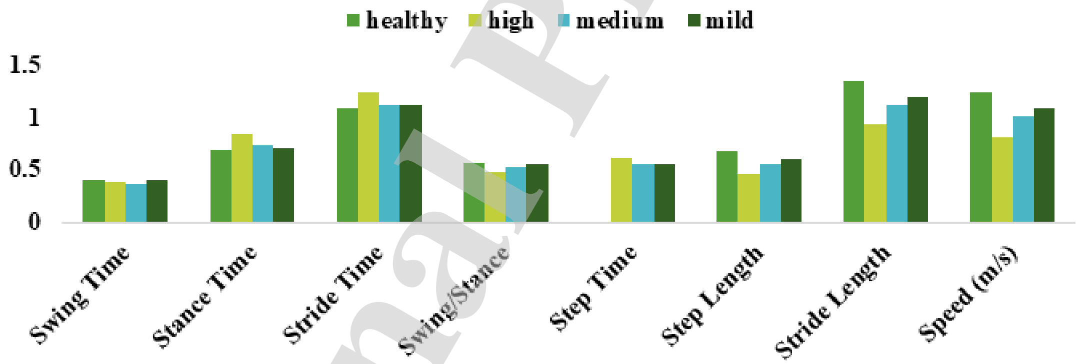


Figure 6: Temporal and spatial features of 4 classes

### 4.3. *Classification algorithms*

#### 4.3.1. *Decision tree*

Decision tree (DT) classifier, which is a supervised machine learning algorithm, is an effective and simple technique that formulates the classification model in the form of a tree structure. Dividing the dataset into smaller subsets in the form of nodes and branches, DT iteratively determines the nonlinear relationship between the input and output of the system. In this study, DT classifies

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the 4 event responses namely healthy, mild, medium and high based on the H & Y scale. Figure 7 shows the tree structure of the DT used for PD classification.

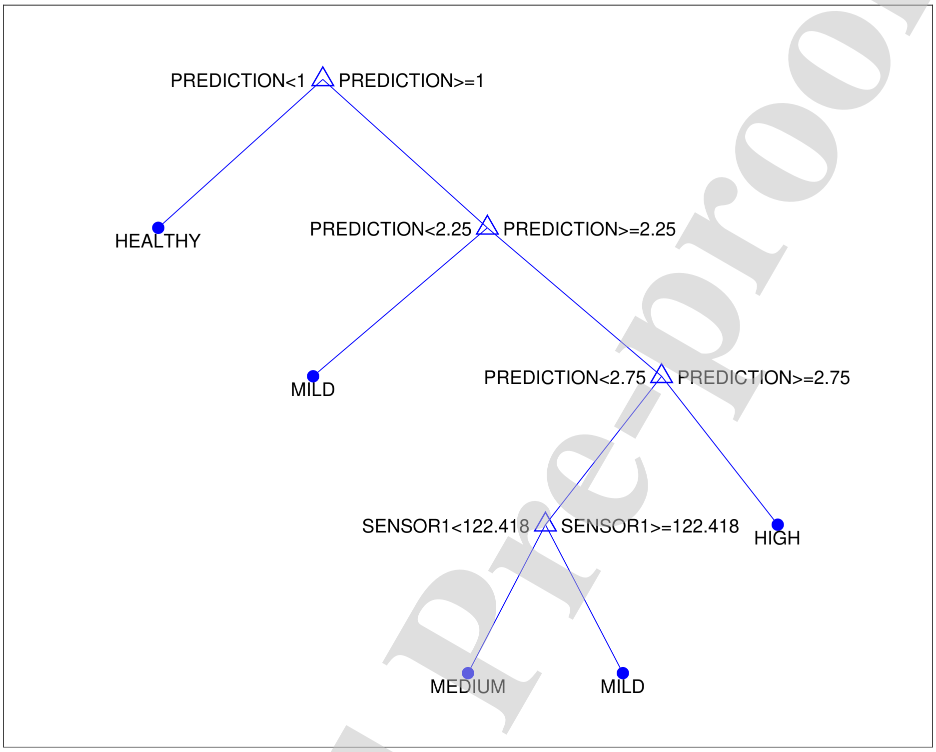


Figure 7: Tree structure of PD classification using DT algorithm

#### 4.3.2. SVM Classifier

SVM, introduced by Boser, Guyon, and Vapnik in 1992, is also a supervised machine learning algorithm, which is highly suited for binary classification problems as well as regression analysis [32]. The prime advantage of SVM is that it can deal with both linear and nonlinear classification problems. Employing kernel methods for mapping the nonlinear data to higher dimensional feature space, SVM utilizes the underlying concept of hyper plane and the margin. SVM finds a hyperplane, also called linear separator, that optimally separates the data into two classes by maximizing the margin between the two classes.

#### *4.3.3. Ensemble Classifier*

Ensemble methods combine several decision tree classifiers that learn the same target function and yield better classification performance by fusing the aggregate of the prediction from each classifier. One of the key advantages of EC is that fusing the outputs of multiple classifiers minimizes the risk of a poorly performing classifier. In this study, we used bagged tree EC which utilizes several subsets of data from training sample selected randomly with replacement to train the decision trees. Creating a base model for each subset, bagged tree EC makes each model to learn from training set in parallel. Consequently, we get the ensemble of different models and the average of all the predictions are utilized to get robust classification result.

#### *4.3.4. Bayes Classifier*

Bayes classifier, which is based on Bayesian theorem, is a statistical classifier that performs the probabilistic prediction of the features and uses the prediction to classify the new test dataset. BC is highly suitable when the input dimensionality is high, and it assumes that the prior probabilistic information about the classes is known [27]. Modeling the probabilistic relationships between the feature vectors and the class variable, BC employs posterior probabilities to identify the class of a test dataset. Given the prior probability function $P(B)$, the likelihood $P\left(\frac{A}{B}\right)$, and the prior probability of predictor $P(A)$, the posterior probability is computed using the Bayes' theorem as follows.

$$P\left(\frac{B}{A}\right) = \frac{P\left(\frac{A}{B}\right) P(B)}{P(A)} \tag{1}$$

## 5. Classifier performance assessment metrics

The performance metrics used for assessing the classifier efficacy are accuracy, sensitivity, specificity, positive predictive value (PPV) (or) precision, negative predictive value (NPV) and F-score. Accuracy, which is the simplest and common measure to assess the classifier, is defined as the degree of correct predictions of the model based on true positive (TP), true negative (TN), false positive (FP) and false negative (FN) values.

$$\text{Accuracy}(\%) = \frac{TN + TP}{TN + TP + FN + FP} * 100\% \tag{2}$$



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Sensitivity measures the proportion of true positives to the total number of actual positives. It indicates how good the diagnostic test can identify the normal (negative) condition.

$$\text{Sensitivity}(\%) = \frac{TP}{TP + FN} * 100\% \tag{3}$$

Specificity characterizes the maximum negative likelihood ratio (NLR) of true negative prediction to actual negatives [33].

$$\text{Specificity}(\%) = \frac{TN}{TN + FP} * 100\% \tag{4}$$

PPV, which is also referred to as precision, represents the proportion of positive predictions to all actual positives. As it conveys how many of diagnostic test positives are true positives, the higher the PPV, the better the results.

$$\text{PPV}(\%) = \frac{TP}{TP + FP} * 100\% \tag{5}$$

NPV indicates the proportion of negative predictions to all actual negatives. It describes how many of test negatives are true negatives.

$$\text{NPV}(\%) = \frac{TN}{TN + FN} * 100\% \tag{6}$$

F-score, which conveys the accuracy of the model, is the weighted harmonic mean of precision and sensitivity.

$$\text{F-score}(\%) = 2 * \frac{\text{Precision*Sensitivity}}{\text{Precision+Sensitivity}} * 100\% \tag{7}$$

## 6. Results and Discussion

The proposed PD stage classification approach is implemented in MATLAB 2018 with the following parameter settings for the classifiers. In DT classifier, Gini's diversity index is chosen as a split criterion with the maximum number of splits being 100 and the surrogate decision splits per node being 10. For SVM, the linear kernel function with box constraint level of 3 and a kernel scale of 4 is configured. In the case of EC, the number of learners is assigned as 30, and the maximum number of splits is set to 165 with the learning rate of 0.1. Moreover, the bagged tree



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ensemble method is chosen for multi-class prediction. In Bayes classifier, the Gaussian kernel function with unbounded support vector is configured, and the multivariate multinomial predictor is set for categorical predictions. The classifier algorithms are run 10 times independently to get an average performance estimate.The training time of the four classifiers namely DT, SVM, EC and BC are 8.06s, 10.04s, 13.09s and 3.19s, respectively.

### 6.1. *Classification performance assessment for statistical features*

Figure 8 shows the scatter plot of each classifier with its respective stage classification based on the SL6 and SL7 sensor data, which have significant fluctuations for all stages of PD. The four stages of dataset include healthy-73, high-10, medium-28 and mild-55. The DT classifier excluding one medium class event, predicts all other events correctly, whereas in SVM, even though all the high events are properly classified, the prediction rate is relatively less in the case of medium and mild events. For instance, 3 medium and 2 mild data variables are unpredicted in the case of SVM. The EC out of the 73 healthy subjects, correctly identifies 72 as healthy subjects and gives one incorrect response. Similarly, in the high, medium and mild events, unpredicted values are 2, 3 and 2, respectively. Among all four classifiers, the BC provides the least prediction results with the incorrect classification in the case of healthy being 4 and medium being 25.

Table V presents the classification accuracy along with the sensitivity and specificity for each stage of the subject for four classifiers. The three key performance metrics for each classifier is illustrated in Figure 9. Moreover, to visualize the cumulative performance of the classifiers for each stage of the subject, the confusion matrix, which is a table layout that gives the summary of the predicted results and the actual results, is shown in Figure 10. From the cumulative performance metrics for each classifier computed from the confusion matrix and presented in Table VI, we can read that in the case of DT, except the medium event, all other events are effectively classified. Even though the misclassification rate in the case of medium event is 4 %, DT classifier offers an overall accuracy of 99.4 %. Moreover, with the least NPV of all classifiers, DT guarantees the lowest misclassification rate for all the four stages considered for classification. SVM which is the second best classifier gives the accuracy of 97.6%, followed by EC and BC with the accuracy of 95.1% and 69.7% respectively.

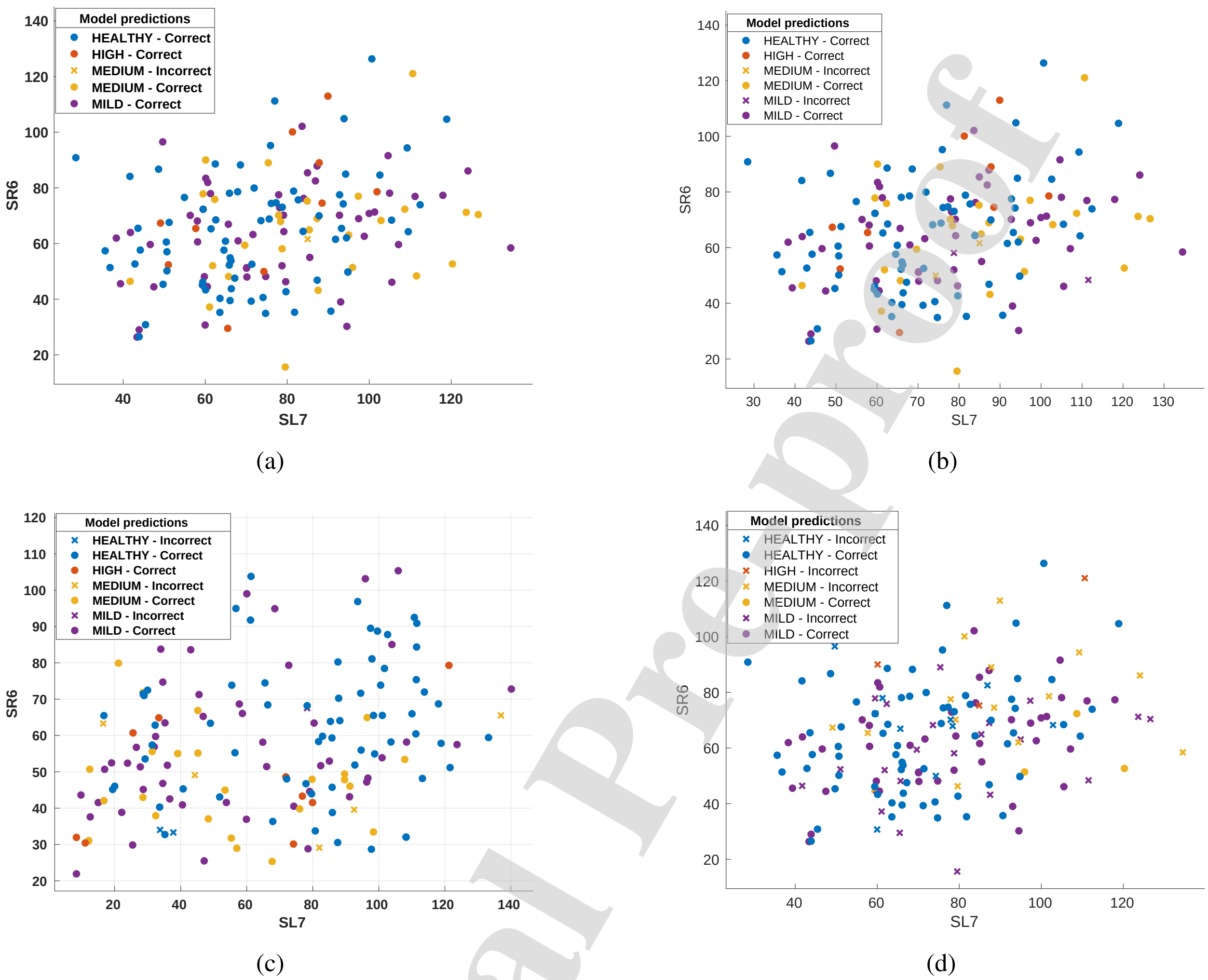


Figure 8: Scatter plot of (a) DT (b) SVM (c) EC (d) BC

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Table V: Classifier performance metrics for four stages based on statistical features

| Classifier | Stage | TN | TP | FN | FP | Accuracy | Sensitivity | Specificity |
|---|---|---|---|---|---|---|---|---|
| DT | Healthy | 73 | 93 | 0 | 0 | 100 | 100 | 100 |
| | High | 10 | 156 | 0 | 0 | 100 | 100 | 100 |
| | Medium | 27 | 138 | 1 | 0 | 99.4 | 100 | 99.3 |
| | Mild | 55 | 110 | 0 | 1 | 99.4 | 98.2 | 100 |
| SVM | Healthy | 73 | 93 | 0 | 0 | 100 | 100 | 100 |
| | High | 9 | 156 | 0 | 1 | 99.4 | 90.0 | 100 |
| | Medium | 25 | 137 | 2 | 2 | 97.6 | 92.6 | 98.6 |
| | Mild | 55 | 108 | 2 | 1 | 98.2 | 98.2 | 98.2 |
| EC | Healthy | 72 | 91 | 2 | 1 | 98.1 | 98.6 | 97.7 |
| | High | 8 | 156 | 0 | 2 | 98.8 | 80.0 | 100 |
| | Medium | 25 | 136 | 3 | 2 | 96.9 | 92.6 | 97.8 |
| | Mild | 53 | 107 | 3 | 3 | 96.3 | 94.6 | 97.2 |
| BC | Healthy | 69 | 83 | 10 | 4 | 91.6 | 94.5 | 89.2 |
| | High | 0 | 153 | 3 | 10 | 92.2 | 0 | 98.1 |
| | Medium | 3 | 123 | 16 | 24 | 76.0 | 11 | 88.5 |
| | Mild | 44 | 89 | 21 | 12 | 80.1 | 78.6 | 80.9 |

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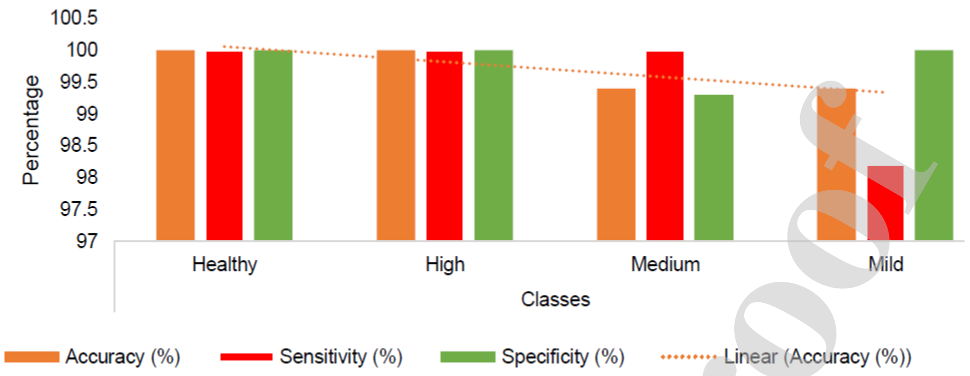
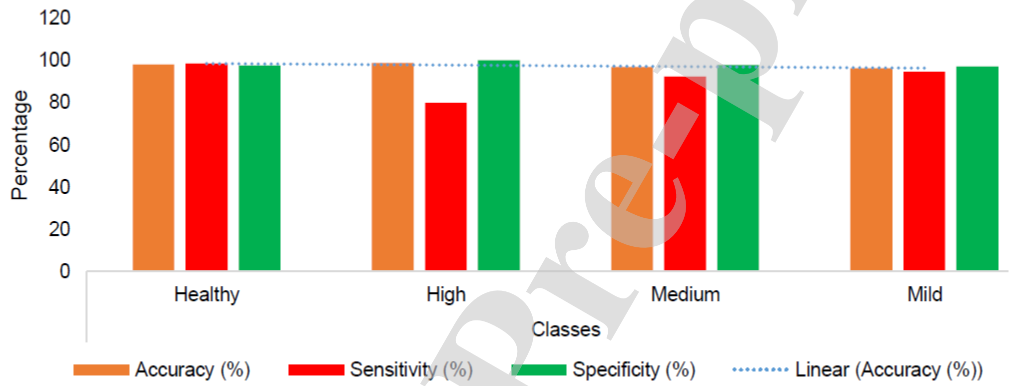
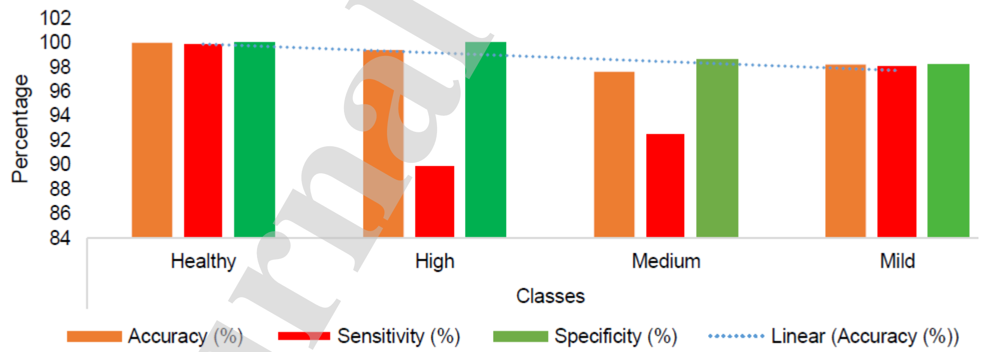


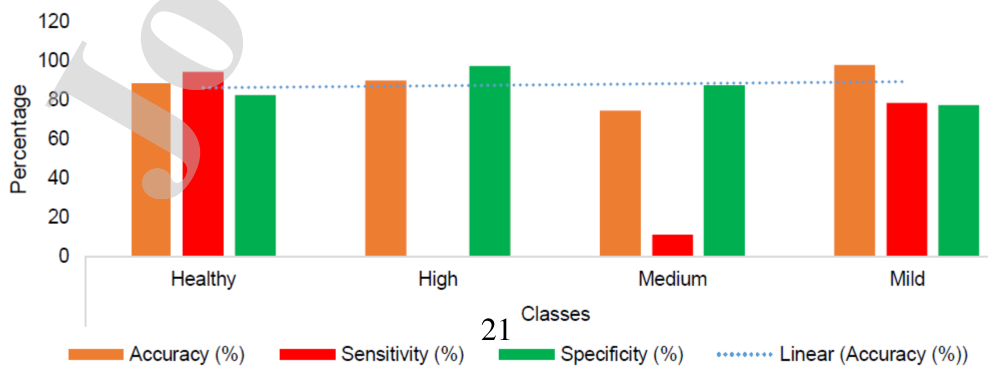

(a)




(b)




(c)




(d)

Figure 9: Performance metrics of classifiers for statistical features (a) DT (b) SVM (c) EC (d) BC

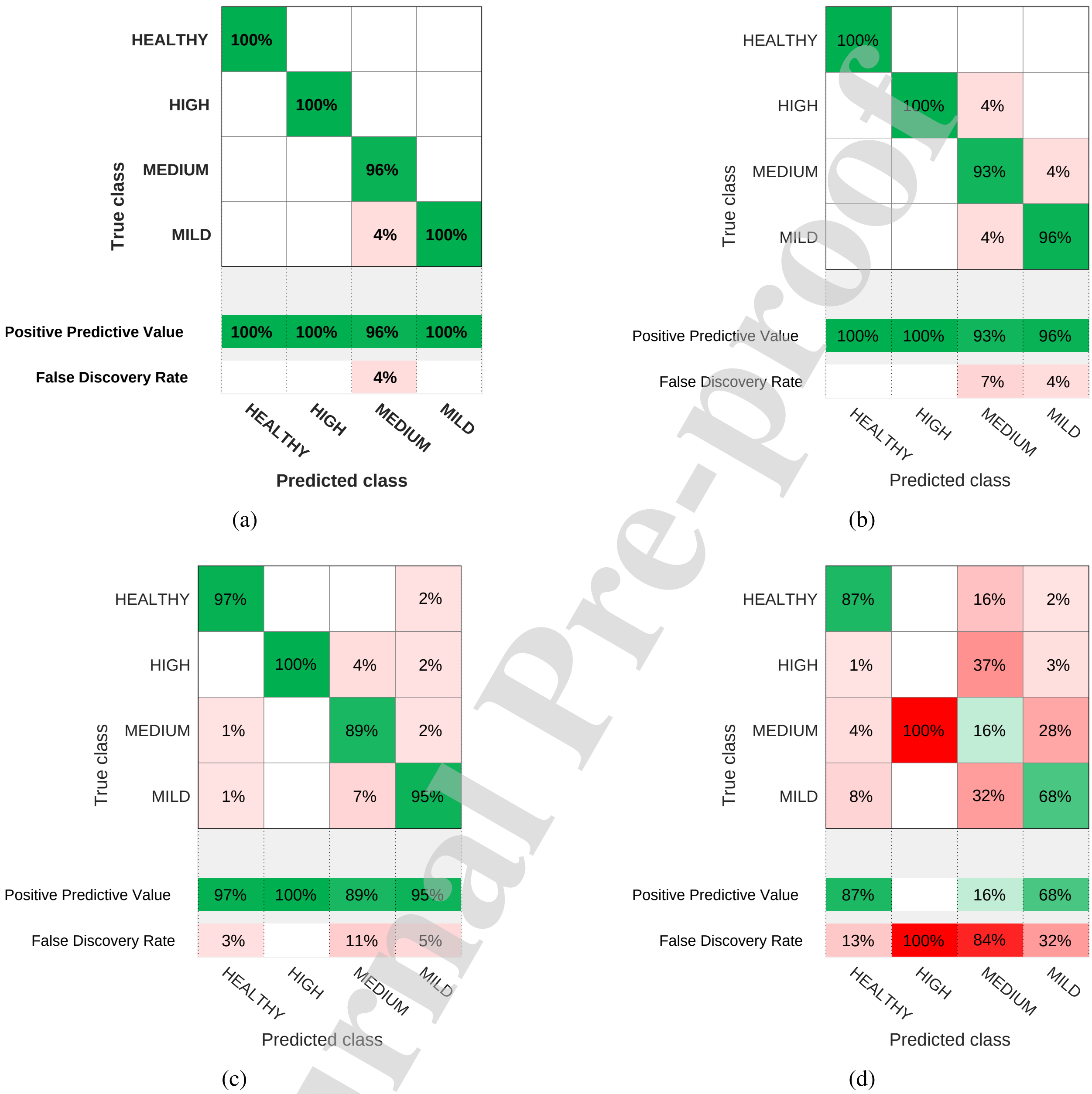



Figure 10: Statistical analysis – Confusion matrix (a) DT (b) SVM (c) EC (d) BC

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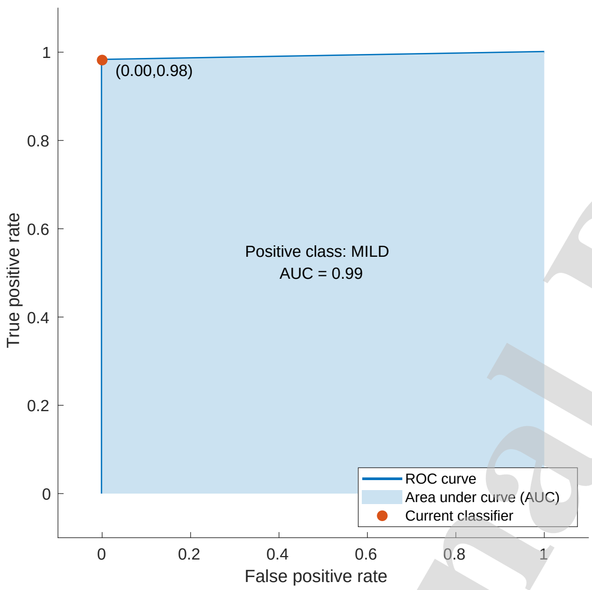
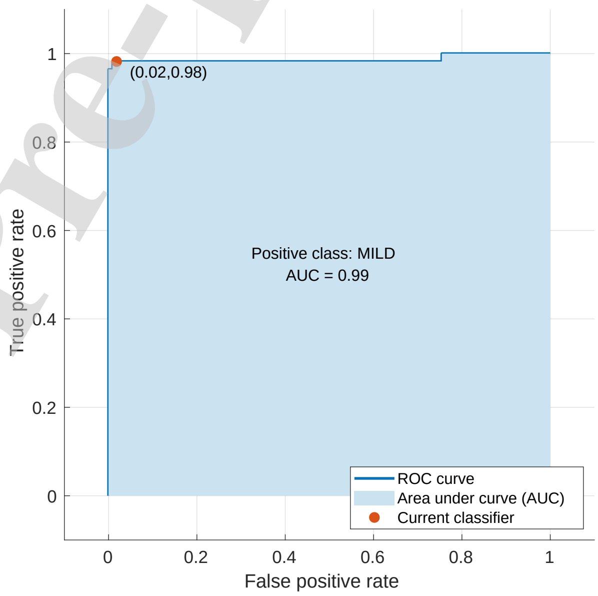
Table VI: Cumulative performance metrics of classifiers for statistical features

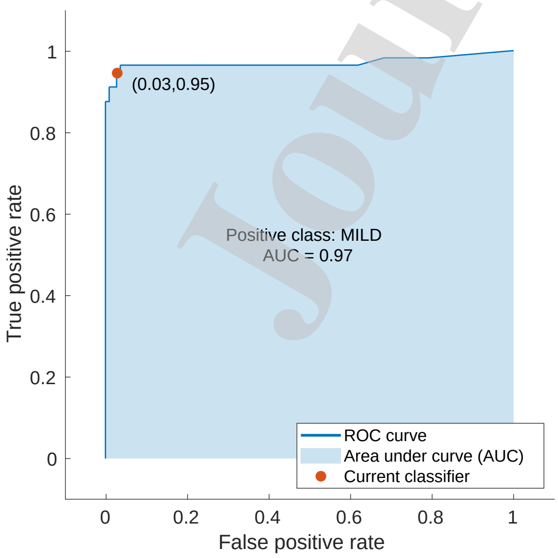
| **Performance Measures** | **DT** | **SVM** | **EC** | **BC** |
|---|---|---|---|---|
| Accuracy (%) | 99.4 | 97.6 | 95.1 | 69.7 |
| Sensitivity (%) | 99.6 | 95.2 | 91.5 | 46.0 |
| Specificity (%) | 99.8 | 99.2 | 98.2 | 89.2 |
| PPV (%) | 99.0 | 97.8 | 94.6 | 67.6 |
| NPV(%) | 1 | 2.2 | 5.4 | 32.3 |
| F-score (%) | 99.25 | 97.3 | 94.6 | 70.7 |




(a)




(b)

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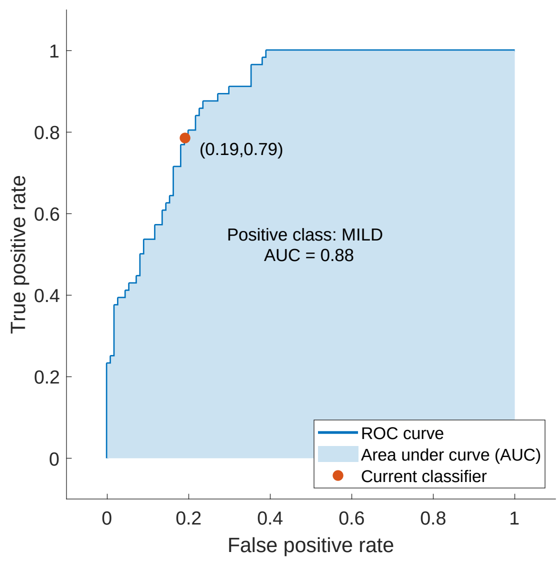


(d)

Figure 11: Statistical analysis – ROC for mild event (a) DT (b) SVM (c) EC (d) BC



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In addition to the accuracy, sensitivity and specificity, an another important measure to assess the quality of classification models is area under curve (AUC) value. The receiver operating characteristic (ROC) curve, which is a plot between the false positive rate (FPR) and true positive rate (TPR), shows the trade-off between sensitivity and specificity. It indicates the degree of separability and conveys how effectively classifiers can distinguish the different classes. The larger the AUC, the better the accuracy of the classifier. For brevity, the AUC of mild event is illustrated in Figure 11. It is to be remarked that even though the classifiers effectively differentiate the stages of the subjects based on statistical features, the major limitation is that the VGRF sensor data may vary based on the weight of the subjects. The statistical analysis of VGRF sometime may not act as a real distinguisher of identifying the gait disorder in PD subjects. Hence, in order to address this issue in the following section we also analyze the walking pattern of the subjects using the kinematic analysis and extract 9 spatiotemporal features from the walking pattern and select the optimal biomarkers to differentiate the healthy subjects and PD subjects.

### 6.2. ***Classifier performance assessment based on kinematic analysis***

Figure 12 shows the kinematic features of healthy and PD subjects for both left and right legs. Comparing the left leg VGRF signals of healthy and PD subjects reveal that the signal amplitude for healthy are almost double as PD subjects amplitude. In addition, the stride phase of PD subjects is relatively less compared to healthy because of the movement disorder. To select optimal feature set from spatiotemporal features, a Pearson rank correlation based feature selection technique is used. Among the 15 spatiotemporal features, 9 features which have significant correlation are chosen. Figure 13, which shows the correlation matrix plot, reveals that the spatiotemporal features have both positive and negative correlation. For instance, the step time and stride time are highly positively correlated. However, the cadence and stance time are highly negatively correlated.

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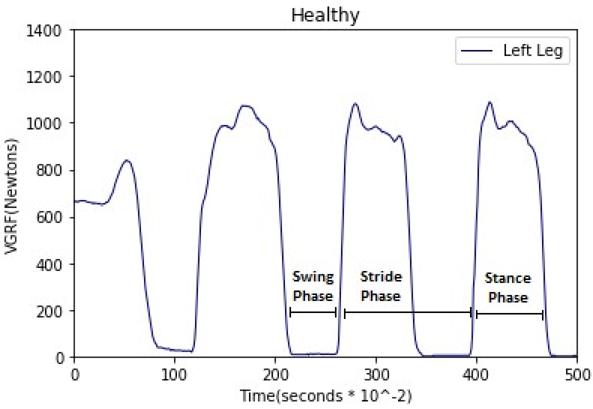
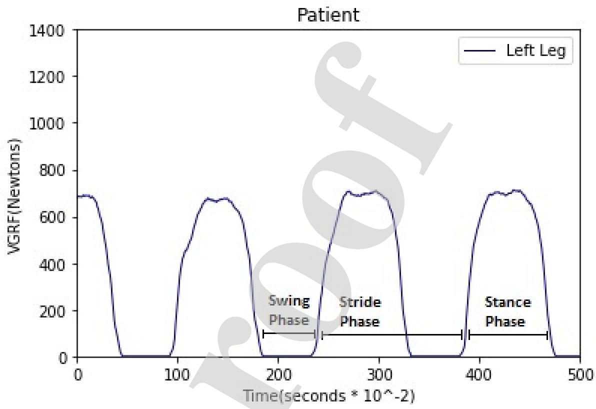
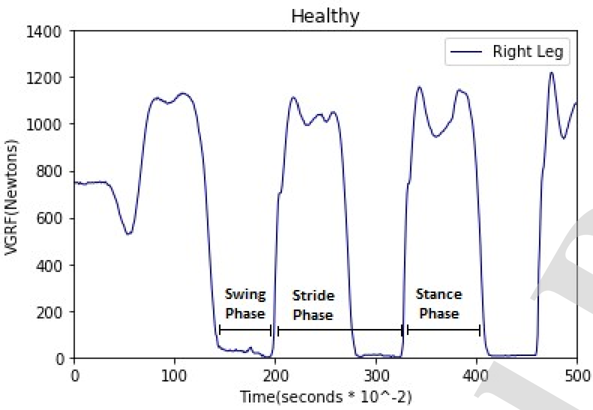


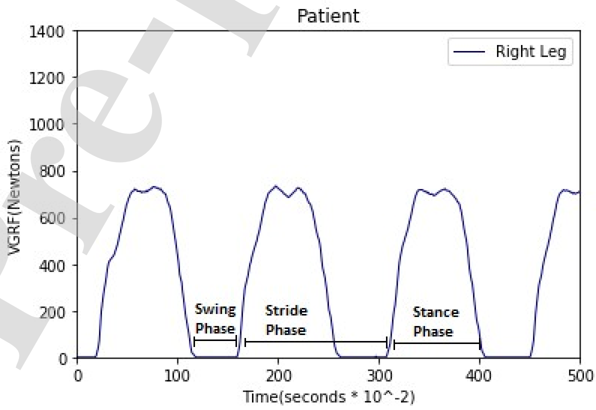

(a) Heathy subject's left leg




(b) PD Patient's left leg




(c) Heathy subject's right leg




(d) PD Patient's right leg

Figure 12: Kinematic parameters of healthy subjects and PD patients

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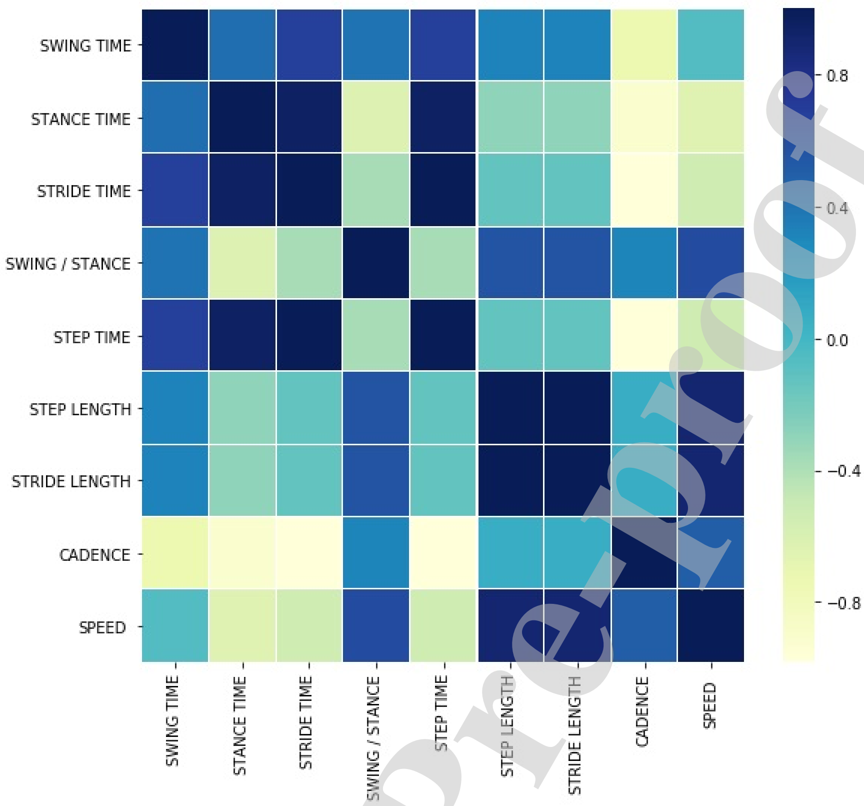


Figure 13: Correlation matrix plot

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Table VII: Kinematic features based performance metrics for four stages of classification

| Classifier | Stage | TN | TP | FN | FP | Accuracy | Sensitivity | Specificity |
|---|---|---|---|---|---|---|---|---|
| DT | Healthy | 73 | 93 | 0 | 0 | 100 | 100 | 100 |
| | High | 10 | 156 | 0 | 0 | 100 | 100 | 100 |
| | Medium | 27 | 139 | 1 | 0 | 99.4 | 100 | 99.3 |
| | Mild | 55 | 111 | 0 | 1 | 99.4 | 98.2 | 100 |
| SVM | Healthy | 73 | 93 | 0 | 0 | 100 | 100 | 100 |
| | High | 10 | 156 | 0 | 0 | 100 | 100 | 100 |
| | Medium | 27 | 138 | 1 | 0 | 99.4 | 100 | 99.3 |
| | Mild | 55 | 110 | 0 | 1 | 99.4 | 98.2 | 100 |
| EC | Healthy | 72 | 91 | 2 | 1 | 98.1 | 98.6 | 97.8 |
| | High | 8 | 156 | 0 | 2 | 98.8 | 80 | 100 |
| | Medium | 25 | 136 | 3 | 2 | 96.9 | 92.6 | 97.8 |
| | Mild | 53 | 107 | 3 | 3 | 96.3 | 94.6 | 97.2 |
| BC | Healthy | 69 | 83 | 10 | 4 | 89.2 | 94.5 | 82.5 |
| | High | 0 | 153 | 3 | 10 | 89.9 | 0 | 97.5 |
| | Medium | 3 | 123 | 16 | 24 | 74.4 | 11 | 87.6 |
| | Mild | 44 | 89 | 21 | 12 | 82.2 | 64.1 | 88.7 |

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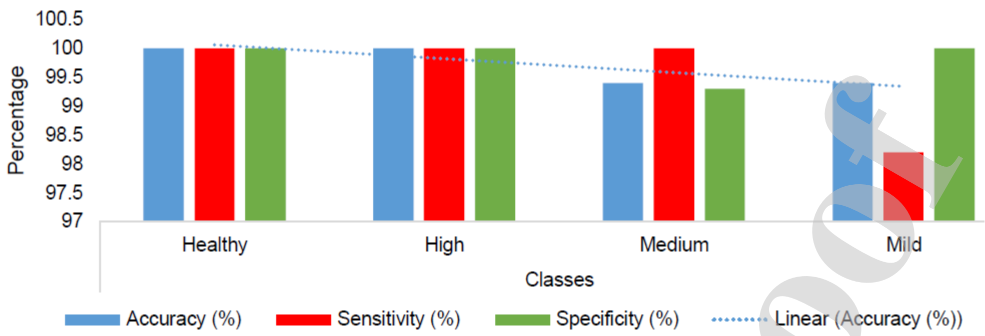
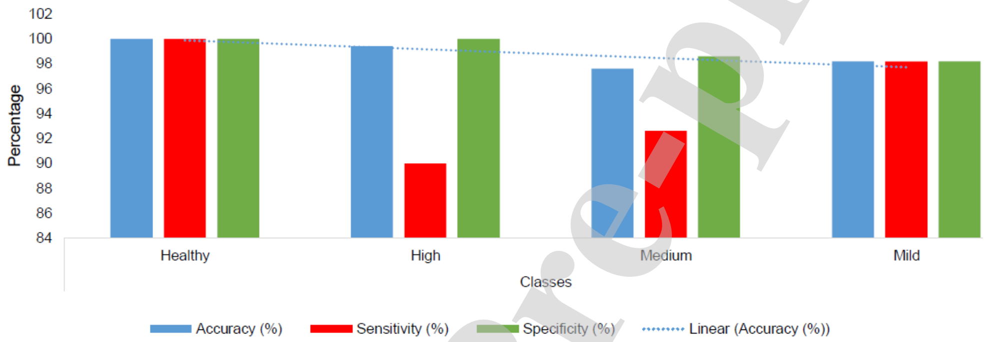



(a)

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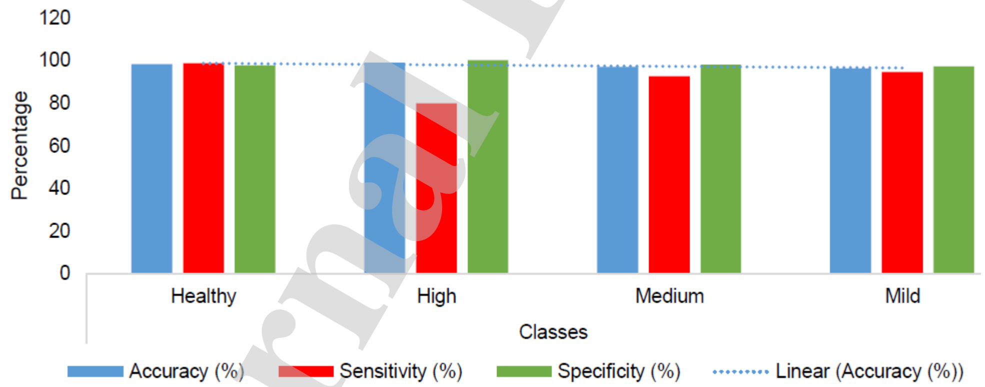


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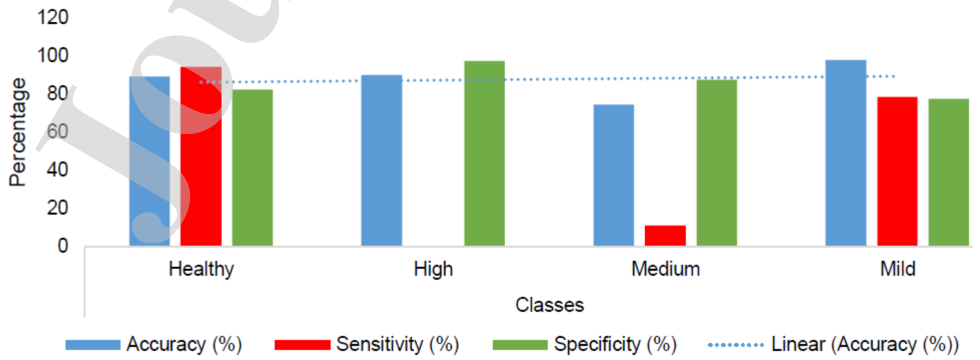




(d)

Figure 14: Performance metrics of classifiers - kinematic analysis (a) DT (b) SVM (c) EC (d) BC

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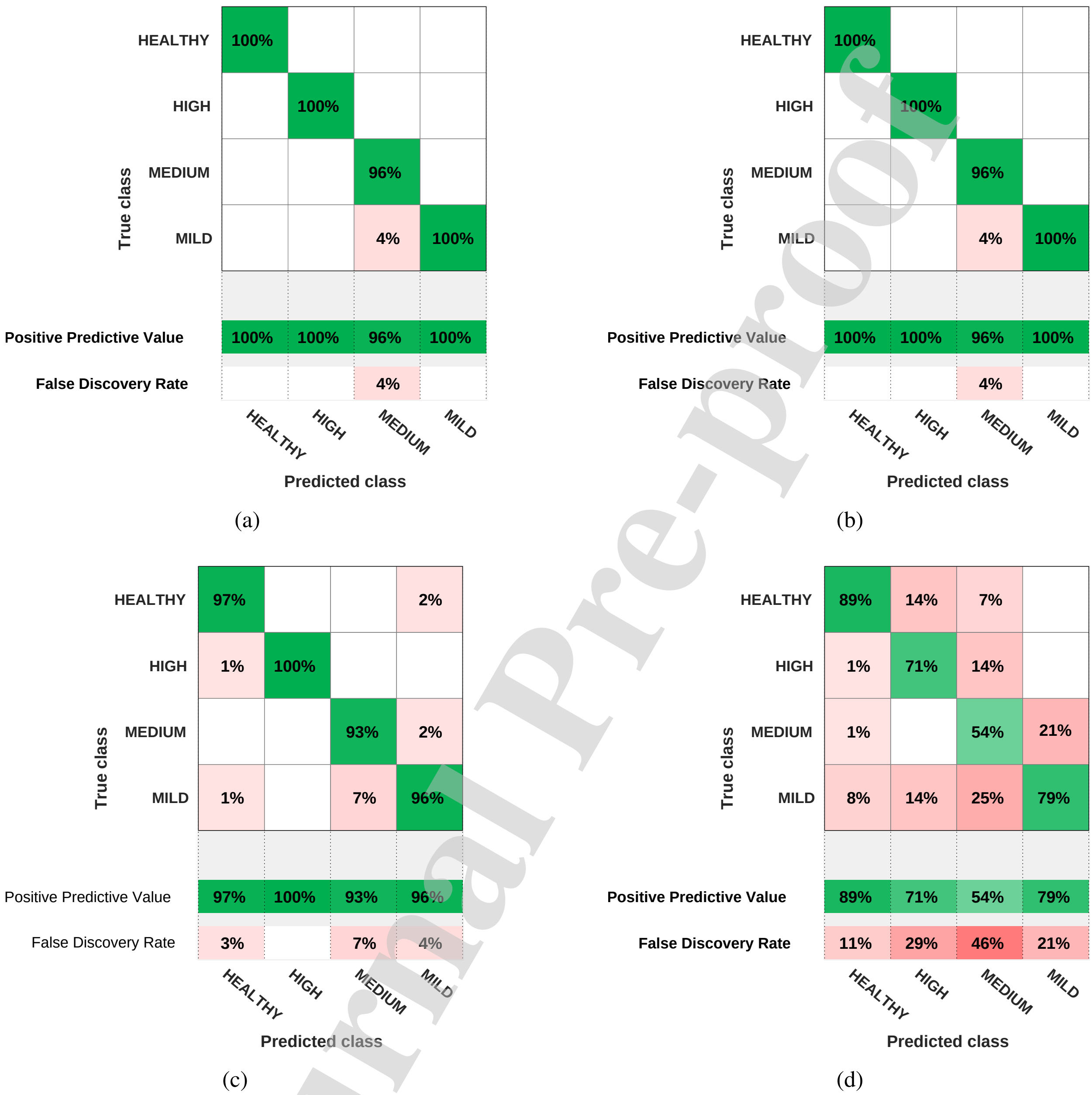


Figure 15: Kinematic analysis - Confusion matrix (a) DT (b) SVM (c) EC (d) BC

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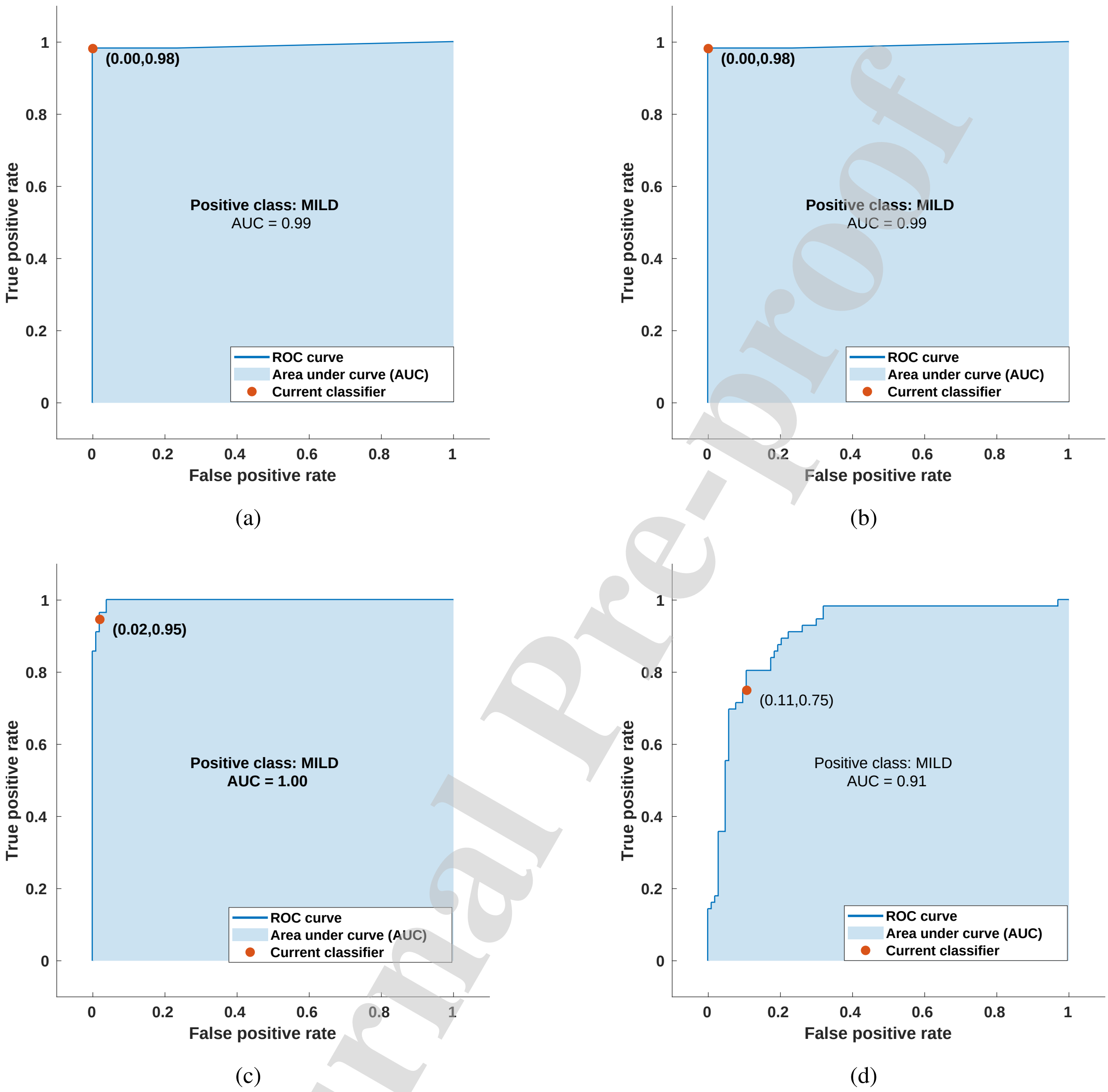


Figure 16: ROC curve for mild event (a) DT (b) SVM (c) EC (d) BC

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Table VIII: Cumulative performance evaluation of classifiers for kinematic features

| **Performance Measures** | **DT** | **SVM** | **EC** | **BC** |
|---|---|---|---|---|
| Accuracy (%) | 99.4 | 99.4 | 95.2 | 69.9 |
| Sensitivity (%) | 99.6 | 99.6 | 91.6 | 42.4 |
| Specificity (%) | 99.8 | 99.8 | 98.2 | 89.1 |
| PPV (%) | 99.0 | 99.0 | 95.3 | 65.7 |
| NPV(%) | 1 | 1 | 4.7 | 34.3 |
| F-score (%) | 99.25 | 99.25 | 93.1 | 72.1 |

Table VII presents the performance metrics for each stage of the classification assessed using the kinematic features, and the three metrics are illustrated in Figure 14. The confusion matrix, shown in Figure 15, highlights that the positive predictive value and false discovery rate of DT and SVM are same for the kinematic features. Excluding the medium stage of PD, all other stages are classified with 100% accuracy. Hence, DT and SVM offer the least misclassification rate of around 4%. However, the misclassification rate of BC, which is around 30%, is the highest at each stage. Table VIII gives the cumulative performance metrics of each classifier for different stages of PD. Moreover, to highlight the misclassification for the mild stage of PD, the ROC and AUC of 4 classifiers are shown in Figure 16. To extend the AUC for a multi-class problem, we utilize the averaging pairwise comparison approach because it gives an overall measure of how effectively each class is differentiated from other classes. Finally, the performance validation of the proposed method to the state-of-the-art methods, which used the same gait dataset from Physionet, is reported in Table IX. It can be noted that the accuracy of the proposed approach is significantly better than those of the other methods and distinguishably the present work utilizes the minimum feature set formed using the spatiotemporal variables. Hence, utilizing the spatiotemporal features can significantly improve the classification accuracy and the proposed approach can serve as a tool for early diagnosis and severity rating of PD based on the clinical symptoms obtained through foot pressure analysis.

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Table IX: Comparison of accuracy of proposed and other reported approaches

| References | Features | Classifiers | Accuracy |
|---|---|---|---|
| Khoury et al., 2019 | Time domain | K-NN, CART, SVM, and GMM | 80-91 % |
| Elkurdi et al., 2018 | Kinematic features | DT, SVM and KNN | 86.19–92.9 % |
| Tavakolian et al., 2017 | Left and right vGRF signals | SVM, KNN, RFDTC | 93.6 % |
| Martinez et al., 2018 | Kinematic features | Linear Discriminant Model | 64.1 % |
| Weizeng et al., 2015 | Kinematic features | RBFNN | 93.1 % |
| Yu Zhang et al., 2016 | Kinematic features | RBFNN | 96.39 % |
| Perumal et al., 2016 | Time and Frequency domain | LDA, SVM and ANN | 86.9 % |
| Prabhu et al., 2018 | Time domain features | SVM, PNN | 96.15 % |
| Proposed framework | Spatiotemporal features | DT | 99.4 % |

### *6.3. Limitations*

Even though the proposed kinematic analysis approach provides better classification accuracy compared to other methods reported on the same dataset, it is also important to mention the limitations of the proposed study. In the current study, we considered only the significant motor symptoms of PD for classification. However, to improve the prediction rate, the non-motor symptoms can also be considered. Moreover, in addition to the gait pattern, the tremor dataset can also be assessed for enhancing the accuracy of the stage classification. Finally, we have considered only the supervised linear classifiers for identifying the stages of PD. Nonetheless, the nonlinear classifiers can be employed for extracting the nonlinear relationships especially when the tremor data is considered along with the gait pattern.

## 7. Conclusions

This paper has presented an automatic gait classification system to diagnose the severity of PD based on the celebrated H & Y scale. To assist the neurologists in PD diagnosis and minimize the misclassification rate in quantifying the stages of PD, a gait classification framework using supervised machine learning algorithms is put forward. Assessing the gait abnormality based on the VRGF dataset using the statistical and kinematic analyses, we have extracted optimal diagnostic biomarkers from spatiotemporal domain and implemented a 10 fold cross validation technique to avoid data overfitting. Moreover, the performance of the classifiers is assessed using the confusion matrix and the ROC curves. Unlike most of the previous machine learning based approaches which are binary classification problem that detects only the presence of PD, the proposed approach can perform a multi-class classification and quantify the stages of PD. Experimental validation substantiates that compared to several state-of-the-art methods, the proposed approach, which uses optimal spatiotemporal domain features, can offer better prediction of stages of PD based on the H& Y scale.

## Acknowledgment

The authors would like to thank the PSG Institute of Medical Science and Research for the support and guidance.

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# Research Highlights

- An automatic gait classification system for identifying the severity level of PD is proposed.
- Four supervised machine learning classifier algorithms: DT, SVM, EC and BC are used to classify the stages of PD.
- Significant biomarkers from temporal and spatial features of gait are extracted.
- Using 10 fold cross validation technique, a PD classification accuracy of 99.4% is obtained.

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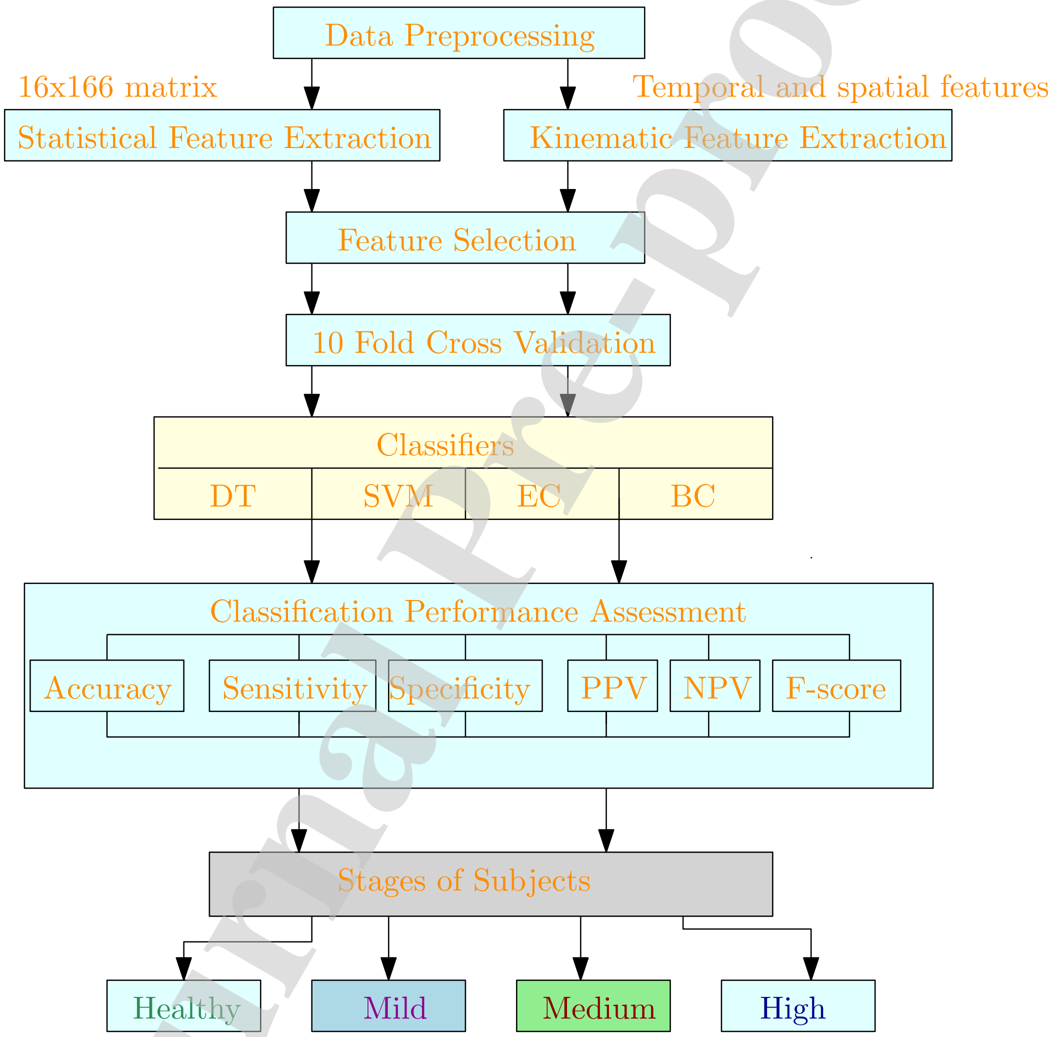
## Graphical Abstract



# CRediT author statement

**Balaji E:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing-Original draft preparation **Brindha D**: Resources, Supervision, Data Curation, **Balakrishnan R**: Visualization, Project administration, Writing-Review &Editing

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**Declaration of interests**

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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